Chapter 5. Basic Oligopoly: Cournot and Bertrand

Monopoly removes strategic interaction: the firm takes market demand as given and chooses its own price or quantity. Oligopoly restores the central problem of industrial organization. A firm's optimal decision depends on what it expects rival firms to do, and each rival is making the same calculation. Cournot and Bertrand are the two canonical models for studying this interaction.

Learning objectives

- Explain Cournot's original two-spring example and the role of quantity commitments.
- Interpret Cournot and Bertrand as reduced-form models that differ in the timing of commitment.
- Write and solve the general Cournot first-order condition and understand best-response functions.
- Derive the Cournot markup-share-elasticity relationship.
- Solve symmetric linear Cournot models and compare monopoly, Cournot, and perfect competition.
- Solve Cournot models with quadratic costs and with asymmetric marginal costs.
- Write the homogeneous-good Bertrand demand-allocation rule and characterize Nash equilibrium.
- Prove the marginal-cost pricing result in the basic Bertrand model.
- Explain why capacity constraints weaken the basic Bertrand conclusion and reconnect price competition to quantity commitments.

5.1 From Cournot's water suppliers to reduced-form oligopoly

Cournot's two springs

Cournot's 1838 analysis began from a concrete homogeneous-product story. He imagined two proprietors controlling two springs of identical quality that supplied the same market.

A useful modern interpretation translates that setting into a spring-water bottling story: firms bottle water and bring it to market. The key economic feature is that output is committed before the market price is known. Once the total quantity arrives, the market-clearing price adjusts to sell it.

Historical note
Cournot did not write an extensive-form "two-stage game" in modern game-theory notation. Treating the spring example as production first and market clearing second is a useful modern interpretation of the economic timing embedded in his quantity-choice model.

A common underlying environment

A useful way to organize oligopoly models is to separate two physical decisions: how much product is made available, and the terms on which it is sold. The reduced-form model depends on which decision is treated as the strategic commitment and which decision is left to adjust afterward.

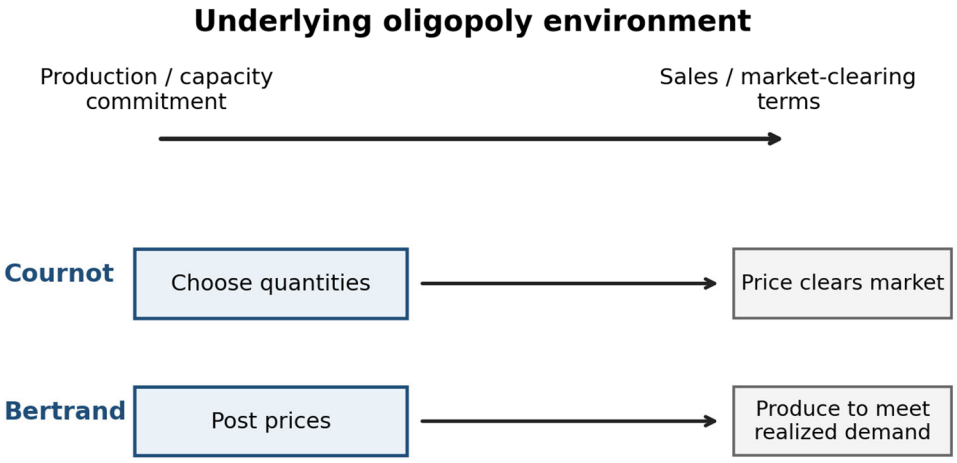


Figure 5.1. Cournot and Bertrand as different reduced forms of an underlying production-and-sales environment.

Model	Strategic commitment	What adjusts afterward	Key residual-demand logic
Cournot	Quantity / production	Market price clears total output	A firm treats rivals' output as fixed when choosing its own output.
Bertrand	Posted price / sales terms	Production expands to meet realized demand	A lower-priced firm can capture demand because output is not precommitted.

The distinction is not merely semantic. In the standard Bertrand model, production follows realized demand: a firm that posts the lowest price is assumed able to supply the customers who arrive. In Cournot, production has already been committed, so cutting price cannot create additional units to sell. Kreps and Scheinkman (1983) formalize this timing logic: in a two-stage game with quantity precommitment followed by price competition, the equilibrium outcome can reproduce the Cournot outcome under suitable assumptions.

Model-selection principle
Ask what is difficult to change quickly. If firms must choose capacity or output before they know the final market price, Cournot may be a useful reduced form. If firms can post prices first and flexibly produce to meet demand, Bertrand may be a useful reduced form.

5.2 Cournot competition: the general model

The strategic environment

There are n firms producing a homogeneous good. Firm i chooses quantity $q_i \geq 0$. Total market output and rivals' output are

$Q = q_1 + \cdots + q_n$, $Q_{-i} = Q - q_i$.

Inverse market demand is P(Q), with $P'(Q)<0$ over the relevant range. Firm i has cost $C_i(q_i)$. Given rivals' output, firm i solves

$\max_{q_i \geq 0} \pi_i(q_i, Q_{-i}) = P(q_i + Q_{-i})q_i - C_i(q_i)$.

For an interior optimum, the first-order condition is

$\partial \pi_i / \partial q_i = P(Q) + P'(Q) q_i - C_i'(q_i) = 0.$

$MR_i(q_i, Q_{-i}) = MC_i(q_i).$

This is the monopoly condition applied to firm i's residual problem. The crucial difference is that the price effect depends on the firm's own quantity q_i, while market price depends on total output Q.

Second-order condition and best responses

A convenient sufficient condition for a local maximum is

$2P'(Q) + P''(Q) q_i - C_i''(q_i) < 0.$

The first-order condition implicitly defines firm i's best-response function $q_i = r_i(Q_{-i})$. A Cournot-Nash equilibrium is a quantity vector q^c such that every firm is choosing a best response to the quantities of all other firms:

$q_i^c = r_i(Q_{-i}^c)$ for every i.

Strategic substitutes

When inverse demand is linear and costs are convex, a rival's larger quantity lowers a firm's best response. More generally, implicit differentiation gives

$dr_i / dQ_{-i} = -[P'(Q) + P''(Q) q_i] / [2P'(Q) + P''(Q) q_i - C_i''(q_i)].$

With linear demand $P'' = 0$, decreasing demand $P' < 0$, and $C_i'' \geq 0$, this derivative is negative. Quantities are strategic substitutes: when rivals expand, firm i optimally contracts.

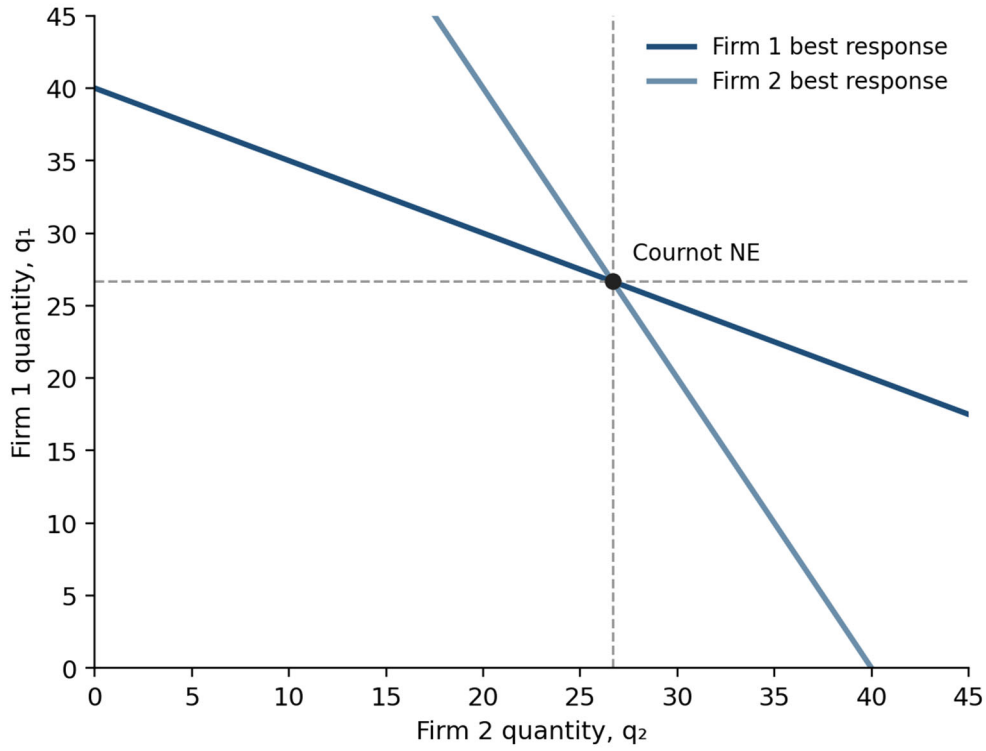


Figure 5.2. Downward-sloping best responses in a linear Cournot duopoly. Their intersection is the Nash equilibrium.

Market share, elasticity, and the Cournot markup

Rearranging the first-order condition yields a useful measure of market power:

$p - MC_i = -P'(Q)q_i$.

Let $s_i=q_i/Q$ be firm i's market share and let $\varepsilon_D<0$ denote the market demand elasticity. Then

$(p - MC_i)/p = s_i / |\varepsilon_D|$.

A Cournot firm has a larger markup when it has a larger market share or when market demand is less elastic. In a symmetric n-firm equilibrium, $s_i=1/n$, so the markup shrinks as the number of symmetric firms rises, holding market elasticity fixed.

Why this matters in IO
The Cournot first-order condition provides a direct bridge from strategic theory to empirical conduct: prices, marginal costs, market shares, and demand elasticities are linked by an equilibrium restriction. This is one reason Cournot remains important well beyond the textbook linear model.

5.3 Cournot competition: linear demand and constant marginal cost

The parametric model

Now specialize to the standard linear model:

$P(Q)=a-bQ$, $b>0$,

$C_i(q_i)=cq_i+F$, $a>c\geq 0$.

The fixed cost F affects profit but not the interior quantity choice. Firm i's profit is

$\pi_i=[a-b(q_i+Q_{-i})]q_i-cq_i-F$.

The first-order condition and best response are

$a-2bq_i-bQ_{-i}-c=0$,

$r_i(Q_{-i})=(a-c)/(2b)-Q_{-i}/2$,

with the understanding that output is truncated at zero if the unconstrained expression is negative.

Symmetric n-firm equilibrium

In a symmetric equilibrium $q_1=\cdots=q_n=q^c$, so $Q_{-i}=(n-1)q^c$. Substituting into the best response gives

$q^c(n) = (a-c)/[(n+1)b]$,

$Q^c(n) = n(a-c)/[(n+1)b]$,

$p^c(n) = (a+nc)/(n+1)$.

When n=1 these expressions reproduce the monopoly solution. As n grows, each individual firm becomes smaller, total output rises, and price falls toward marginal cost.

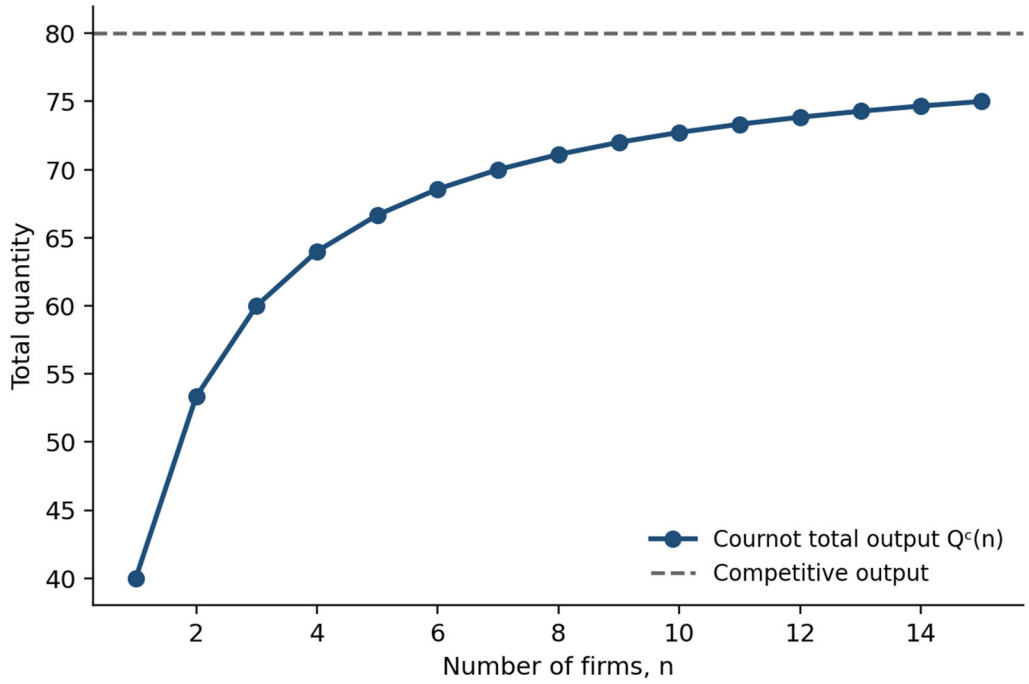


Figure 5.3. With linear demand and constant marginal cost, Cournot total output approaches the competitive output as n increases.

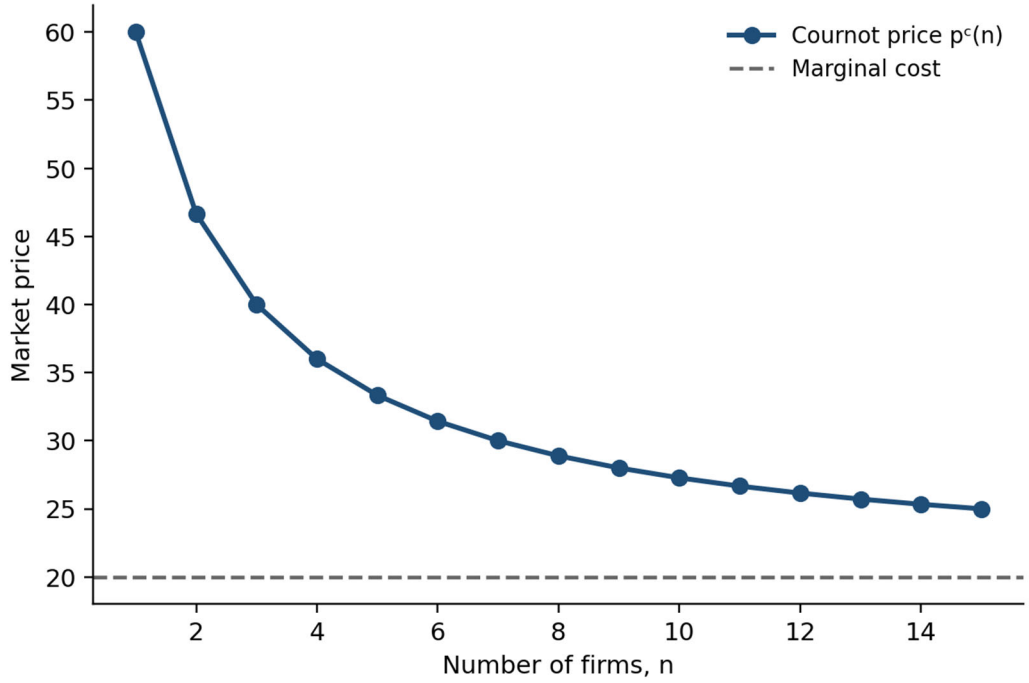


Figure 5.4. The Cournot price approaches marginal cost as the number of firms increases.

Profit and welfare

With F=0, each firm's operating profit is

$\pi_i^c = (p^c - c)q^c = (a-c)^2/[b(n+1)^2]$.

Consumer surplus, total producer surplus, and total surplus are

$CS^c = n^2(a-c)^2 / [2b(n+1)^2]$,

$PS^c = n(a-c)^2 / [b(n+1)^2]$,

$TS^c = (a-c)^2/[2b] \cdot n(n+2)/(n+1)^2$.

The efficient quantity is $Q^* = (a-c)/b$ and efficient total surplus is $(a-c)^2/(2b)$. Hence

$DWL^c = (a-c)^2 / [2b(n+1)^2]$.

The welfare loss falls rapidly as n rises. The limiting result $Q^c \to Q^*$ and $p^c \to c$ formalizes the benchmark ordering

$Q^m < Q_2^c < Q_3^c < \cdots < Q_\infty^c = Q^{Pc}$,

$p^m > p_2^c > p_3^c > \cdots > p_\infty^c = MC$.

Worked example: two firms

Consider $P(Q)=180-2Q$, marginal cost $c=20$, and fixed cost $F=10$. With $n=2$,

$q_1^c = q_2^c = (180-20)/(3 \cdot 2) = 80/3$,

$Q^c = 160/3$, $p^c = 220/3$.

Each firm's profit is

$\pi_i^c = (220/3-20)(80/3)-10 = 12{,}710/9 \approx 1{,}412.22$.

Calculation discipline
Do not impose $q_1=q_2$ before taking a firm's derivative. A firm chooses q_i while treating q_j as fixed. Symmetry is imposed only after the best-response condition has been derived. This is the same warning important in every symmetric Cournot calculation.

A useful benchmark comparison

Using the same demand $P(Q)=100-Q$ and constant marginal cost $c=20$, the monopoly, Cournot-duopoly, and Bertrand outcomes can be compared directly:

Market structure	Quantity	Price	Total operating profit
Monopoly	40	60	1,600
Cournot duopoly	$160/3 \approx 53.33$	$140/3 \approx 46.67$	$2 \cdot (80/3)^2 \approx 1{,}422.22$
Bertrand / perfect competition	80	20	0

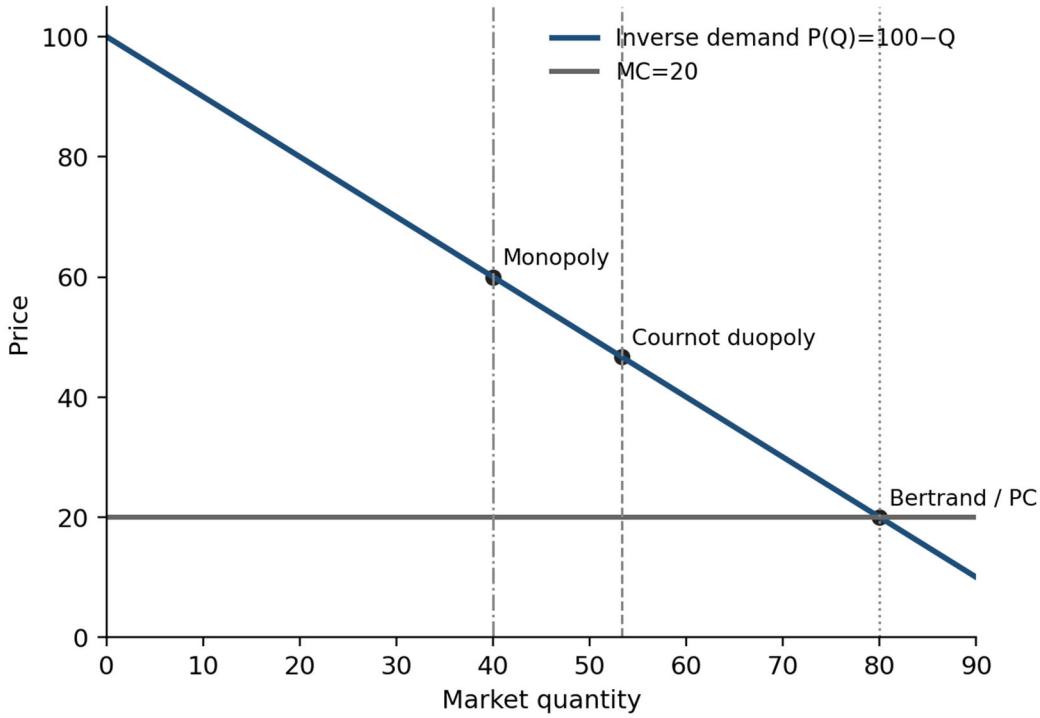


Figure 5.5. The same primitives can generate very different outcomes depending on the strategic variable and timing.

5.4 Cournot with convex and asymmetric costs

Quadratic costs

The notes next consider convex costs, which make marginal cost rise with a firm's own output:

$P(Q)=a-bQ$, $C_i(q_i)=d\,q_i^2/2 + F$, $d\geq 0$.

The first-order condition is

$a-2bq_i-bQ_{-i}-dq_i=0$,

so the best response is

$r_i(Q_{-i})=a/(2b+d)-[b/(2b+d)]Q_{-i}$.

For n symmetric firms,

$q^c = a/[b(n+1)+d]$,

$Q^c = na/[b(n+1)+d]$,

$p^c = a(b+d)/[b(n+1)+d]$.

Convex cost dampens each firm's expansion relative to the constant-marginal-cost case. As the number of firms rises, production is also spread across more firms, which lowers the industry marginal cost of supplying a given total quantity.

Parametric model: $P(Q)=200-Q$ and $C_i(q_i)=cq_i^2/2$

For the special case b=1 and d=c, The best response and symmetric n-firm outcome are

$r_i(Q_{-i}) = [200-Q_{-i}]/(2+c)$,

$q^c(n)=200/(n+1+c)$,

$p^c(n)=200(1+c)/(n+1+c)$.

Using the cost function as stated, firm profit is

$\pi_i^c = p^c q^c - (c/2)(q^c)^2 = (1+c/2)[200/(n+1+c)]^2$.

For $c=1$ and $n=2$, $q^c=50$, $Q^c=100$, and $p^c=100$. Then

$\pi_i^c=100(50)-50^2/2=3{,}750$,

$CS^c=5{,}000$, $TS^c=12{,}500$.

Asymmetric constant marginal costs

Now consider two firms with

$P(Q)=A-Q$, $C_1(q_1)=c_1q_1$, $C_2(q_2)=c_2q_2$,

where firm 1 is the lower-cost firm: $A>c_2>c_1>0$. The interior best responses are

$r_1(q_2)=(A-c_1)/2-q_2/2$,

$r_2(q_1)=(A-c_2)/2-q_1/2$.

If both firms are active, solving the two equations gives

$q_1^c=(A-2c_1+c_2)/3$,

$q_2^c=(A-2c_2+c_1)/3$,

$Q^c=(2A-c_1-c_2)/3$, $p^c=(A+c_1+c_2)/3$.

The lower-cost firm produces more. A one-unit reduction in c_1 raises q_1 by 2/3, lowers q_2 by 1/3, raises total output by 1/3, and lowers price by 1/3. Strategic substitution therefore transmits one firm's cost advantage to its rival's output choice.

$CS^c=(2A-c_1-c_2)^2/18$.

Because the first-order condition implies $p^c-c_i=q_i^c$ in this unit-slope demand model, operating profits are especially simple:

$\pi_1^c=(q_1^c)^2$, $\pi_2^c=(q_2^c)^2$.

Interior-solution check
The condition $A>c_2>c_1$ alone does not guarantee that the higher-cost firm produces a positive quantity. The interior formulas require $A-2c_2+c_1>0$. If that condition fails, the constrained best response of firm 2 is zero and the active-firm set must be reconsidered.

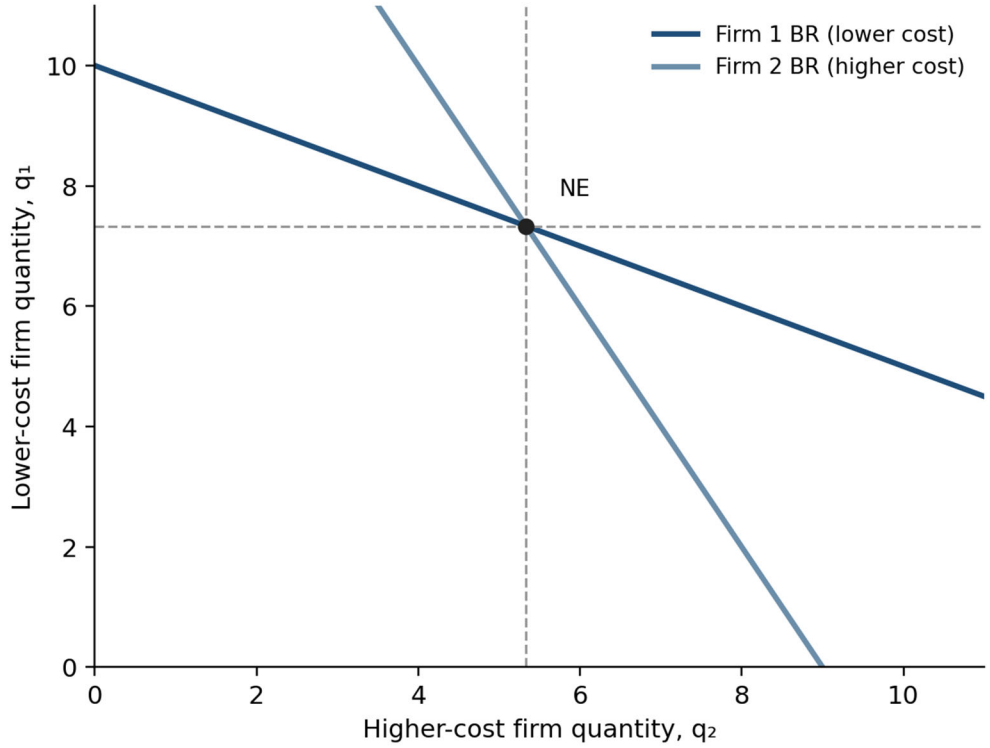


Figure 5.6. Asymmetric Cournot best responses. The lower-cost firm has the larger equilibrium quantity.

Empirical connection: electricity generation

Cournot is often used as a conduct benchmark when output and capacity are difficult to adjust quickly. A prominent example is wholesale electricity. Puller (2007), using detailed firm-level data from California's deregulated electricity market, found that generator conduct was fairly consistent with a Cournot pricing game for much of the 1998–2000 sample. Borenstein, Bushnell, and Wolak (2002) separately documented substantial departures from competitive pricing during high-demand periods. These studies do not imply that generators literally play the simple textbook game; rather, they show why a quantity-based strategic benchmark can be empirically useful in capacity-constrained industries.

5.5 Bertrand competition: the general model

Prices are the strategic variables

Bertrand's 1883 critique of Cournot asked what happens when firms compete in prices rather than quantities. In the basic homogeneous-good model, firms post prices before they know how much they will sell, and production then adjusts to satisfy the demand generated by those prices. The ability to expand output after observing demand is the crucial contrast with Cournot.

Let market demand be $D(p)$, with $D'(p)<0$. For the core benchmark assume $n\geq 2$ firms, identical constant marginal cost c, no capacity constraint, and no product differentiation. Let $p_{-i}\hat{}\min$ be the lowest price among firm i's rivals.

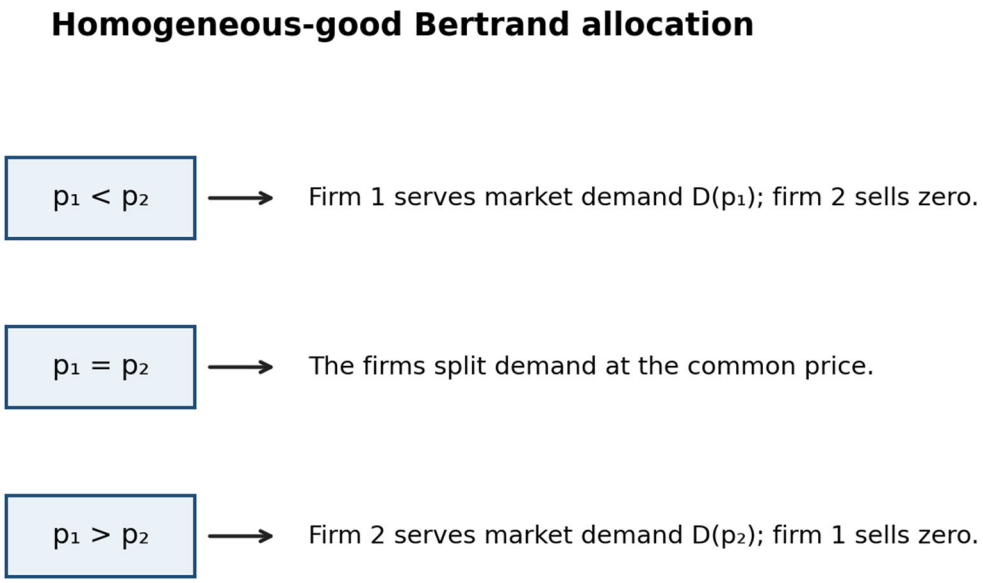


Figure 5.7. Demand allocation in the homogeneous-good Bertrand model.

If m firms tie for the lowest price, they split demand equally under the standard symmetric tie-breaking rule. Firm i's operating profit is

$\pi_i(p_i,p_{-i}) = (p_i - c)\, q_i(p_i,p_{-i})$.

Unlike Cournot profit, this objective is generally discontinuous at rivals' prices because a tiny undercut can change a firm's quantity from a fraction of market demand to all market demand.

The Bertrand equilibrium theorem

Bertrand equilibrium with identical costs
With n≥2 homogeneous-good firms, identical constant marginal cost c, unlimited capacity, and continuous prices, a pure-strategy Nash equilibrium has a minimum price equal to c and at least two firms charging c. In a duopoly this implies the unique equilibrium $p_1=p_2=c$.

The logic is easiest to see in three steps.

1. No firm can price below marginal cost in equilibrium. A firm that makes positive sales at p<c earns a negative margin and can instead choose a price that yields zero sales and zero operating profit.

2. The lowest market price cannot be above marginal cost. If one firm is uniquely lowest at d>c, it can raise price slightly while remaining below the next-lowest rival. If several firms tie at d>c, one of them can undercut by a very small amount and capture the entire market. Either way, the candidate profile is not stable.

3. At least two firms must charge c. If exactly one firm charged c while every rival charged more, that firm could raise its price slightly above c while remaining the unique lowest-price seller and earn positive profit.

Conversely, if at least two firms charge c and all other firms charge prices weakly above c, no firm has a profitable unilateral deviation. A firm charging c cannot profitably undercut because that creates a negative margin; raising its price loses demand and still yields zero profit. A high-price firm cannot profitably undercut c.

Welfare implication

At p=c, market output is D(c), exactly the quantity at which willingness to pay equals marginal cost. Thus the basic Bertrand outcome is allocatively efficient, consumer surplus is maximized subject to the technology, and operating profit is zero. This is why the result is often called the Bertrand paradox: two firms can be enough to produce the competitive outcome.

What the paradox is really telling us
The Bertrand result is not a claim that two real-world firms always eliminate market power. It is a statement about a very demanding collection of assumptions: homogeneous products, identical costs, unconstrained production, perfect price observability, and the ability of the lowest-priced firm to serve the market. The value of the model is that it makes those assumptions transparent.

5.6 Bertrand competition: parametric examples and equilibrium proofs

Parametric model: D(p)=100−p

Consider

$D(p)=100-p$, $n\geq 2$, $MC_i=c$.

With two identical firms and zero fixed cost, the equilibrium is

$p_1^b=p_2^b=c$.

At the tie, each firm serves one-half of market demand,

$q_1^b=q_2^b=(100-c)/2$,

but each firm earns zero operating profit because price equals marginal cost.

A clean duopoly proof

To verify that (c,c) is a Nash equilibrium, hold firm 2 at $p_2=c$. If firm 1 chooses $p_1<c$, then

$\pi_1=(100-p_1)(p_1-c)<0$.

If firm 1 chooses $p_1>c$, it sells zero and earns zero. At $p_1=c$ it also earns zero. Thus no deviation produces a strictly higher payoff.

To prove uniqueness in a duopoly, suppose some equilibrium differs from (c,c). An asymmetric pair $p_1>p_2$ cannot be stable: if $p_2\leq c$, the low-price firm can avoid a nonpositive margin, while if $p_2>c$, the high-price firm can undercut p_2 and earn positive profit. A symmetric pair $p_1=p_2=d<c$ gives negative profit and is dominated by selling nothing; a symmetric pair $d>c$ can be profitably undercut by an arbitrarily small amount. Hence only (c,c) survives.

More than two firms

With $n>2$ identical firms, $p_i=c$ for every i is an equilibrium, but it is not the only price vector. The full equilibrium set requires that at least two firms charge c and all remaining firms charge prices at least c. This qualification is important: the market outcome is unique—price c and quantity D(c)—even though the vector of posted prices need not be unique.

5.7 Why Cournot and Bertrand differ

The sharp difference between Cournot and Bertrand comes from the residual opportunity available after a rival changes strategy. Under Cournot, rival output is already committed. If a rival produces more, firm i's residual demand shrinks and firm i responds by producing less. Under Bertrand, output is not committed in advance. If a rival charges a high price, firm i can undercut slightly and expand production to serve the customers it attracts.

Feature	Cournot	Bertrand
Strategic variable	Quantity	Price
Timing interpretation	Production before market clearing	Price before realized production
Typical best-response relation	Strategic substitutes in quantities	Discontinuous undercutting incentives
Two identical firms, homogeneous good	Price above MC	Price equals MC
Effect of more firms	Price falls gradually toward MC	Two active firms already suffice for p=MC under basic assumptions
Critical assumption	Output commitment is meaningful	Lowest-price firm can serve demand

The models are therefore not rival mathematical tricks. They are different hypotheses about the institution and timing of competition.

5.8 Capacity constraints and the Edgeworth correction

Why unlimited capacity is decisive

Edgeworth's critique of the stark Bertrand outcome focuses on a simple issue: a low-price firm may not be able to serve the entire market. Once capacity is limited, the high-price firm can retain residual demand, so the winner-take-all logic of a tiny price cut is weakened.

Consider market demand of 10 units, marginal cost 2, and capacity 7 for each firm. Firm 1's sales are

$q_1=7$ if $p_1<p_2$; $q_1=5$ if $p_1=p_2$; $q_1=3$ if $p_1>p_2$.

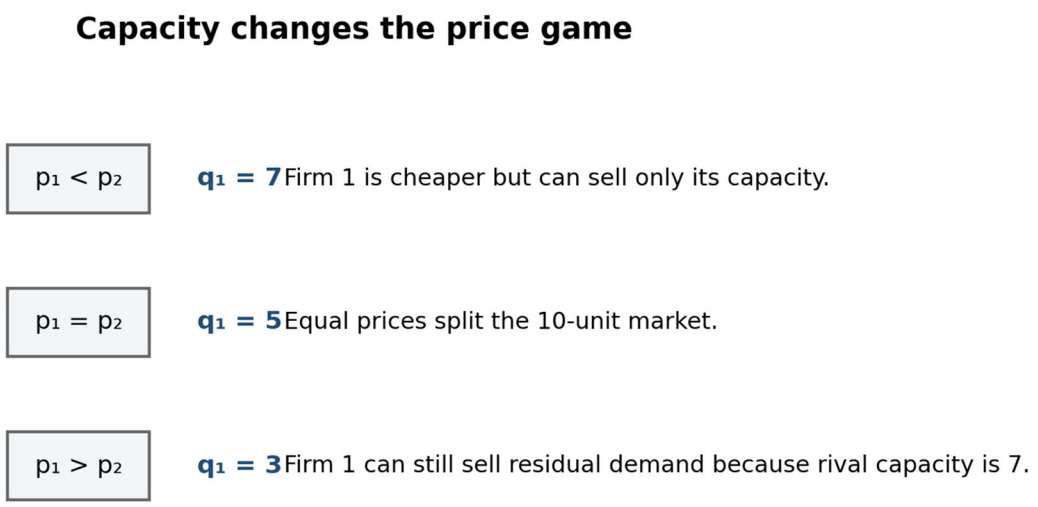


Figure 5.8. With capacity 7 and market demand 10, even the high-price firm can sell residual demand.

The qualitative Edgeworth lesson: capacity constraints can destroy the simple p=MC conclusion and may generate multiple equilibria, cycles, or nonexistence of a pure-strategy price equilibrium depending on the exact rationing rule and demand system.

Capacity first, price second

Levitan and Shubik (1972) analyze price duopoly with capacity constraints, and Kreps and Scheinkman (1983) make the timing link especially clear. Their two-stage logic is

- Stage 1: firms commit quantities or capacities.
- Stage 2: after observing those commitments, firms compete in prices.

Under suitable assumptions, backward induction yields the Cournot quantity outcome. This provides a microfoundation for the statement that Cournot can be interpreted as a reduced form of “production before sales.” It also explains why the Bertrand result depends not just on choosing price as a strategic variable but on allowing production to follow realized demand without a binding prior commitment.

5.9 Empirical interpretation and model choice

Cournot as a reduced-form conduct model

Quantity competition is often a useful approximation in industries where capacity or production schedules are sticky relative to market clearing. Electricity generation is a leading example because storage is limited, production constraints matter, and demand can be highly inelastic in the short run. Puller (2007) found California generator behavior fairly consistent with a Cournot benchmark for much of his sample. Borenstein, Bushnell, and Wolak (2002) found substantial market-power effects during high-demand periods. These papers illustrate how the Cournot first-order condition can be used as an empirical benchmark even when the literal institutional details are richer than the textbook game.

Bertrand as a benchmark for price competition

Price posting is central in many retail markets. Gasoline stations, for example, visibly post prices and consumers choose where to buy. Hastings (2004) studies how changes in local retail gasoline structure affected price competition in Southern California. But gasoline is differentiated by location, brand, convenience, and contractual arrangements, so the pure homogeneous-good Bertrand model is best viewed as a benchmark. Later chapters on product differentiation replace the winner-take-all demand rule with smoother firm-level demands and generate positive markups even when prices are the strategic variables.

Industry example: capacity commitments in the DVD-by-mail market

The basic Bertrand model assumes that a low-price firm can expand output to serve the customers attracted by its price. Belleflamme and Peitz (2015, Case 3.3) use the DVD-by-mail industry to show why this assumption can be questionable. Demand for newly released films was much higher than demand for older titles, so firms such as Netflix and its rivals had to decide in advance how many physical copies of popular releases to hold. Extra inventory was costly, and an unexpectedly successful low price could leave a firm unable to serve all of the demand it attracted.

This is exactly the economic environment in which the distinction between price competition and quantity or capacity commitment matters. Capacity is chosen before the price-setting stage, so the later pricing game is constrained by an earlier physical decision. The example therefore provides an industry interpretation of the Edgeworth and Kreps–Scheinkman material: when firms cannot costlessly expand supply after prices are posted, the stark homogeneous-good Bertrand result can be a poor reduced form.

Source text: Belleflamme and Peitz (2015), Case 3.3, “When capacity choices condition pricing decisions in the DVD-by-mail industry.”

5.10 Summary and study guide

- Cournot’s original spring-water example models independent suppliers of a homogeneous product choosing quantities that jointly determine market price.
- A useful modern interpretation treats Cournot and Bertrand as reduced forms of different timing: production before market clearing versus price before realized production.
- In general Cournot competition, each firm satisfies $P(Q)+P'(Q)q_i=MC_i$ and chooses a best response to rivals’ output.
- Under standard conditions, Cournot quantities are strategic substitutes.
- The Cournot markup satisfies $(p-MC_i)/p=s_i/|\varepsilon_D|$.
- With linear demand and constant marginal cost, symmetric Cournot output rises and price falls as the number of firms increases, converging to perfect competition.
- Asymmetric costs shift best responses: lower-cost firms expand, higher-cost firms contract, and market price falls when a firm becomes more efficient.
- In basic homogeneous-good Bertrand competition with identical constant marginal costs and unlimited capacity, the market price equals marginal cost.
- For $n>2$ Bertrand firms, equilibrium requires at least two firms at marginal cost; the market outcome is unique even though every posted price need not be identical.
- Capacity constraints weaken the Bertrand undercutting logic and reconnect price competition to prior quantity commitments.

Common mistakes to avoid

- Imposing symmetry before deriving a Cournot best response.

- Using market quantity Q as if an individual Cournot firm controlled all of it.
- Forgetting to truncate a best response at zero when the interior formula is negative.
- Confusing an individual firm's Cournot profit with total producer surplus.
- Claiming that every n-firm Bertrand equilibrium requires every firm to price at c; only at least two firms must do so.
- Using the Bertrand p=MC result when a low-price firm cannot supply the market.
- Treating Cournot versus Bertrand as merely a choice of algebra rather than a choice about timing and market institutions.

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